Luvion Whitepaper

An Algorithm-Agnostic High-Threshold MPC Signing Framework with Dynamic Committee Rotation and Self-Healing Network

Document Type: Technical Whitepaper
Version: 1.0.0
Status: Public Draft for External Review
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Disclaimer

This whitepaper describes protocol architecture, implementation direction, and benchmark references.
It does **not** represent production security guarantees.
No third-party audit is complete at publication time.
Any deployment handling real assets must pass independent security review and staged production hardening.

Revision History

Version	Date	Notes
1.0.0	2026-04-15	First publishable format draft

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10. References

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1. Executive Summary

Luvion is an open-source threshold MPC signing framework designed for institutional self-custody systems requiring stronger compromise resistance than low-threshold static deployments.

The framework combines: - high-threshold operation (`22-of-33` by default), - dynamic committee rotation, - self-healing share maintenance under node churn, - algorithm-agnostic backend design across classical and post-quantum signing paths.

A key engineering contribution is **Lagrange pre-multiplication** in threshold ML-DSA aggregation, addressing hint-weight overflow and restoring practical signing validity under FIPS 204 parameter bounds.

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2. Problem Landscape

Production MPC deployments often show four recurring gaps:

1. **Low threshold concentration:** compromise cost remains too low in 2-of-3 and 3-of-5 patterns.
2. **Static committee exposure:** fixed long-lived key-share holders enable prolonged targeting.
3. **Recovery fragility:** node churn frequently triggers expensive full-DKG cycles or manual operations.
4. **Missing PQ migration path:** many systems remain tied to ECDSA-only operational assumptions.

Luvion targets all four in one protocol family.

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3. Protocol Architecture

3.1 System Model

- Candidate pool size: `N >= 100`
- Active committee size per epoch: `n = 33`
- Signing threshold: `t = 22`
- Byzantine tolerance target: `f = 11` (`t = 2f + 1`)

No node can reconstruct the global signing secret alone.

3.2 Backend Abstraction

Luvion separates orchestration and arithmetic backend through a uniform interface:

- `keygen(params, threshold, parties)`
- `sign_partial(share, message, participant_set)`
- `aggregate(partial_sigs, public_key)`
- `verify(signature, message, public_key)`

This enables backend substitution: - classical backend (`secp256k1` family), - post-quantum backend (`ML-DSA-65` family).

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4. Threshold ML-DSA Technical Core

4.1 Parameter Profile (ML-DSA-65)

Parameter	Value
q	8,380,417
n	256
k, l	6, 5
gamma1	131,072
gamma2	95,232
tau	49
eta	4
beta	196
omega	55

4.2 Three-Round Signing Outline

Round 1: Commitment

Each participant samples bounded masking vector `y_i`, computes `w_i = A * y_i`, and broadcasts commitment.

Round 2: Lagrange-Scaled Response

Each participant computes `lambda_i` on active set `S` and outputs: - `z_i = (y_i + c * s1_i) * lambda_i` - `cs2_i = (c * s2_i) * lambda_i`

Round 3: Aggregation

Coordinator computes: - `z = sum(z_i)` - `cs2 = sum(cs2_i)`

4.3 Correctness Intuition

Using Lagrange interpolation identity over `t-of-n` shares:

`sum(lambda_i * s_i) = s`

thus:

`sum(lambda_i * (c * s2_i)) = c * s2`

This restores algebraic equivalence to single-signer form and keeps hint weight within specification bounds.

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5. Dynamic Committee and Network Resilience

5.1 Dynamic Selection

Committee rotation is epoch-based and verifiable, reducing attacker pre-targeting value of static membership.

5.2 Self-Healing Operations

- proactive share resharing on node replacement,
- incremental DKG for partial membership changes,
- coordinator failover through view-change logic,
- protective signing lock under quorum loss.

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6. Security Model and Assumptions

6.1 Threat Model Snapshot

Threat	Vector	Mitigation
Single-node compromise	malware / insider	no full key at node level
Sustained collusion	long-term targeting	dynamic committee rotation
Node churn / partition	availability degradation	resharing + quorum lock
Coordinator DoS	round disruption	BFT-style view change
Quantum pressure on signing	future cryptanalytic shift	ML-DSA path

6.2 Assumptions

- epoch-level honest threshold is maintained,
- network delay is bounded for liveness,
- standard hardness assumptions hold for selected backend,
- hash primitives modeled in ROM.

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7. Implementation Status and Benchmarks

Current implementation tracks modular Rust components for:

- threshold signing,
- lagrange arithmetic,
- polynomial ring operations,
- DKG and VSS families,
- network orchestration paths.

Benchmarks are representative of controlled 33-node test environments; WAN behavior requires separate operational trials.

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8. Risks, Limitations, and Mitigation Roadmap

8.1 Current Risks

- audit not completed,
- some production-hardening components still in staged maturity,
- PQ committee-randomness migration not fully closed.

8.2 Mitigation Priorities

1. Independent external audit completion.
2. Cryptographic hardening and protocol review closure.
3. WAN validation and failure-injection testing.
4. Multi-team code ownership and independent reviewer model.

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9. Conclusion

Luvion positions high-threshold MPC as a practical institutional architecture by combining threshold hardness, dynamic unpredictability, and self-healing operation.
Its threshold ML-DSA aggregation method with Lagrange pre-multiplication is a key step toward practical post-quantum-capable threshold signing systems.

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10. References

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